# HW 03: Data Management

**Wrangling your data into analyzable form**

## Purpose

By now you should know what variables you want to use, and you've looked over the codebook enough now that you have an idea of some potential problems that you will encounter. This assignment uses your chosen research data, and the variables that you chose in the last assignment when you created a personal research codebook. You will thoughtfully review the variables you are interested in, document which ones will need changed and how.

All raw data needs to stay raw, and all changes need to be documented. You will create a script/code file that will make all changes to the data in a programatically and reproducible way. You will create a single code file that imports your raw data, performs some data cleaning steps, and saves out an analysis ready data set that you will use throughout the semester.

You are **not** expected to have completed data management for every one of your variables under consideration by the submission date. I want to see a VERY good effort has been made (raw data read in, at least 2 quant and 2 cat variables dealt with, analysis data saved out.)

Check the rubric in Canvas for more grade specific details.

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## Instructions

1. Download this template starter script and save it into your `math615/script` folder. Rename the `dataset` portion of the file name to your data name (e.g. `dm_addhealth`)
2. After you load the data, restrict the variables to only the ones you are investigating using the `select` function from the `dplyr` (or `tidyverse`) package.
3. Explore and clean at least 2 categorical and 2 numeric variables. One by one, check each variable for necessary adjustments. Complete each of the following steps for each variable.
    - First explain in English what the variable name is and what it measures.

- Then examine the variable using the `table()`, `summary()` and `hist()` functions.
- Identify the data type of the variable using the `class()` or `typeof()` functions. Ask (and answer) the question "Does this match with the intended data type?"
- Recode the data as necessary
- Always confirm your recodes worked as intended by creating another table or summary.

4. Export the resulting data set to your `data` folder as `datasetname_clean.Rdata`, e.g. `addhealth_clean.Rdata` using the `save()` function

Reference lec03b-data_prep and ASCN Chapter 1 on how you can explore your data and start to make data edits using R. Not sure how to start? Check out the walk through video in Canvas.

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## Submission instructions

- You will submit your code file to Canvas. This must run on my computer.
  - What comes out is what I grade.
  - Render your script file to PDF before you submit to ensure that it works and looks the way you expect it to. Fix any code that wraps off the page.
  - You will upload your data set to a Box link contained in Canvas.

If there are questions I may ask you to demonstrate your code to me in person.

> **!** Ensure Reproducibility
>
> To ensure reproducibility between research partners, AND my self, you must do the following:
>
> - Be using an R Project for this class
> - Your data must be in your `data` folder, and your scripts must be your `scripts` folder
> - Use the `here::here()` method of accessing your data

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## Example

> **Tip**
>
> Here is an example using Quarto. Everything you see below this line comes is written in the `.qmd` script itself. This is an example of literate programming where you write what you are going to do, do it in code, and then write what you have learned after each step.

### Setup

```
# Load libraries
library(tidyverse)

# Read in the data
raw <- read.delim(here::here("data/Depress.txt"), sep="\t",  header=TRUE) %>%
  # convert all variable names to lowercase
  janitor::clean_names()

# Select only the variables that I am interested in
mydata  <- raw %>% select(age, marital, cesd, health)
```

### General Health

The variable `health` records a persons perceived general health as being either Excellent, Good, Fair or Poor. This is considered an ordinal categorical variable.

```
table(mydata$health)
```

```
  1   2   3   4
130 115  35  14
```

```
class(mydata$health)
```

```
[1] "integer"
```

The variable `health` currently is an integer with numeric values 1-4, but the codebook states that this is a categorical variable where 1=Excellent, 2=Good, 3=Fair, 4=Poor. So I need to convert this numeric variable to a factor variable. There are no values outside the 1-4 range, such as a -9 that codes for missing data so I do not need to make any further adjustments (You want to code out missing before you convert variables to factors)

```
mydata$health_cat <- factor(mydata$health, labels=c("Excellent", "Good", "Fair", "Poor"))
```

I will confirm that the recode worked by making a two-way `table`

```
table(mydata$health, mydata$health_cat, useNA="always")
```

```
      Excellent Good Fair Poor <NA>
  1         130    0    0    0    0
  2           0  115    0    0    0
  3           0    0   35    0    0
  4           0    0    0   14    0
  <NA>        0    0    0    0    0
```

This shows that all 1's are now 'excellent', 4's are now 'poor' and so forth.

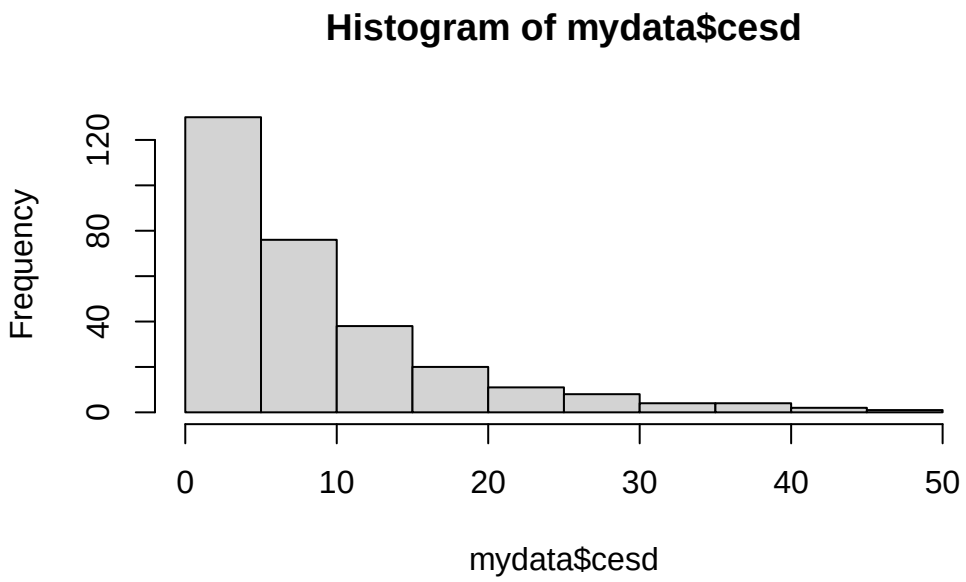
### CESD

The numeric variable `CESD` represents the depression index scale, which is a sum of 20 component variables. A high score indicates a person who is depressed, with 16 being the typical cutoff for creating a binary indicator of depression.

```
summary(mydata$cesd)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   3.000   7.000   8.884  12.000  47.000
```

```
hist(mydata$cesd)
```




There are no values outside of the expected range, and no missing values.

### Indicator of Depression

Create the binary indicator of depression using CESD

```
mydata$depressed <- ifelse(mydata$cesd > 16, "depressed", "not depressed")
table(mydata$depressed, useNA="always")
```

```
    depressed not depressed          <NA>
           45            249             0
```

No missing values were accidentally generated,

### Export cleaned data set

Keep only selected variables and save to an external file

```r
clean <- mydata %>% select(age, marital, cesd, depressed, health_cat)
save(clean, file=here::here("data/depression_clean.Rdata"))
```